HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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ABSTRACT

Detecting tiny objects in high-resolution aerial and maritime imagery is severely challenged by background redundancy and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer from severe semantic collapse on micro targets, causing irrecoverable coarse-stage pruning. We propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a dual-evidence framework featuring semantic-spectral routing, coverage-constrained spatial optimization, and staged lightweight downstream detection. HESOD couples semantic objectness with channel-pooled spectral saliency at constant channel dimension under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. To promote recall-safe routing, an object-level soft-coverage loss directly optimizes multi-cell joint instance coverage across ground-truth objects. To recover downstream efficiency, an inverted residual decoupled head (ISPP) is fine-tuned via selector-first staged training. Extensive experiments show that HESOD improves $\mathrm{mAP@.5}$ by 0.9 pp and 2.3 pp (boosting total recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, with $> 27\%$ fewer parameters and negligible FLOPs overhead, establishing an effective accuracy-efficiency co-design suitable for resource-constrained deployment.

***Index Terms*—** Small object detection, selective computation, spectral saliency, coverage optimization, UAV

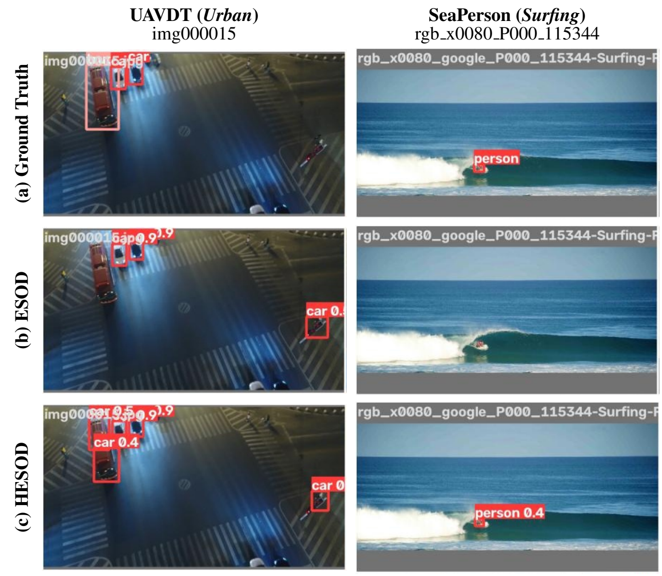

Fig. 1. Qualitative comparison on UAVDT and SeaPerson.

1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ($< 16 \times 16$ px) amid massive uniform backgrounds ($> 70\%$–80%) [4, 5]. Evaluating standard dense deep detectors across multi-megapixel grids (1280×1280 to 2048×2048) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* dynamically routes feature processing strictly to prospective foreground regions [4, 6, 7]. Notably, ESOD [4] predicts an early spatial priority heatmap directly after a shallow stem, slicing features into candidate bounding patches (AdaSlicer) and bypassing background across downstream stages.

Crucially, spatial selection operates under an *asymmetric error cost*: retaining a false-positive background patch merely introduces minor local compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative* that permanently removes the instance from all downstream stages [4]. Despite notable efficiency gains, existing spatial selectors exhibit two critical vulnerabilities: (i) *Evidence Fragility*: relying solely on single semantic objectness causes severe degradation on micro targets after downsampling [8]; and (ii) *Coverage Deficit*: standard per-pixel binary cross-entropy (BCE) supervises grid cells independently, failing to guarantee joint multi-cell coverage across an instance's physical spatial extent and often pruning faint targets (Fig. 1).

To resolve these challenges, we observe that tiny targets retain prominent high-frequency spatial gradients and spectral saliency even when deep semantic responses fade [8]. We propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a unified dual-evidence framework integrating semantic-spectral routing, coverage-constrained spatial optimization, and staged compute recovery (Fig. 2). Specifically, our key contributions are:

1. **Semantic-Spectral Dual Selection**: We couple semantic objectness with channel-pooled spectral saliency at strictly $\mathcal{O}(1)$ width complexity, extracting directional edge gradients to rescue degraded micro targets from semantic collapse.
2. **Evidence Preservation and Soft Coverage**: We formulate a monotonic elementwise max-fusion rule and an object-level soft-coverage loss to directly optimize joint multi-cell instance recall under irreversible pruning.
3. **Lightweight Staged Downstream Detection**: We design an inverted residual decoupled head (ISPP) fine-tuned via selector-first staged training, cutting parameters by **27.5%** and GFLOPs by **16.9%** at **106.1 FPS** with peak detection accuracy.

Source code and weights will be available upon acceptance.

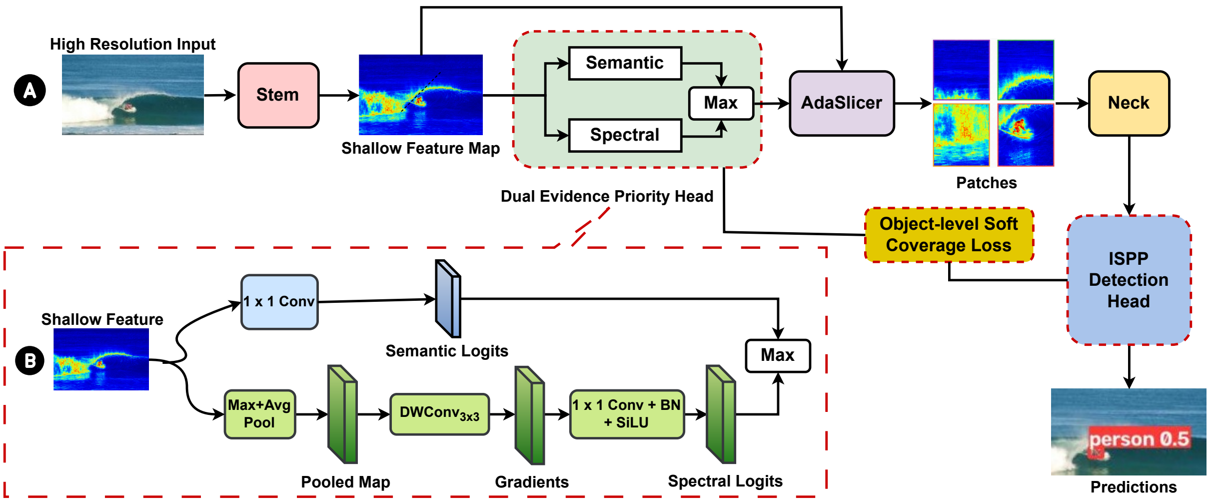

Fig. 2. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Dashed boxes highlight our proposed contributions over ESOD.

2. RELATED WORK

2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ($> 70\%$–80%) [1–3, 9]. Classical approaches utilize density-guided cropping (ClusDet [10], DMNet [11]) or scale-adaptive label assignment (NWD [12], RFLA [13]). Recently, ResBiDet [14] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [5], QueryDet [6], CEASC [7]). Notably, ESOD [4] predicts an early-stage spatial priority heatmap to extract bounding patches (AdaSlicer). However, ESOD relies exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

2.3. Spectral Saliency and Frequency Methods

Frequency-domain representations offer rich structural priors for micro targets, which retain sharp high-frequency edge gradients even when photometric semantics fade [8]. While SET [8] applies dense frequency transforms across multi-scale pyramids with prohibitive latency, HESOD repurposes spectral saliency as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional Laplacian and Sobel gradient filtering at strictly $\mathcal{O}(1)$ width complexity.

3. PROPOSED METHOD

The overall architecture of HESOD is illustrated in Fig. 2(a). Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a shallow stem extracts dense feature map $\mathbf{F} \in \mathbb{R}^{H_s\times W_s\times C}$. The *Dual-Evidence Spatial Selector* then coordinates two stages: (i) a Dual-Evidence Priority Head fuses high-level semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) to generate a continuous spatial priority map $\mathbf{S} \in [0,1]^{H_s\times W_s}$; and (ii) an AdaSlicer routing module dynamically clusters candidate foreground cells into bounding patches $\mathcal{P}$ directly from $\mathbf{F}$. Downstream backbone stages, neck, and lightweight ISPP decoupled heads evaluate only these retained patches, supervised by an object-level soft-coverage loss and trained via a selector-first staged protocol (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class-specific logits via a 1×1 convolution: $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1\times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$. In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \,\|\, \frac{1}{C}\sum_{c=1}^{C} \mathbf{F}_{:,:,c}\right] \in \mathbb{R}^{H_s\times W_s\times 2}, \quad (1)$$

where $[\cdot \,\|\, \cdot]$ denotes channel concatenation. Compressing $\mathbf{F}$ to 2 channels retains both the peak local edge activation and background energy profile, while guaranteeing that subsequent spatial filtering complexity remains strictly $\mathcal{O}(H_s W_s)$ with constant channel dimension, regardless of backbone width C.

Next, a learnable depthwise 3×3 convolution bank—initialized with discrete Laplacian (isotropic second-order edge transitions) and directional horizontal/vertical Sobel operators—filters $\mathbf{F}_{\text{pooled}}$ into 6 structural channels. A 1×1 projection layer maps the filtered channels to dimension N_c, yielding spectral logits $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$.

To prevent confidence dilution on weak targets, evidence fusion enforces *evidence preservation* via a parameter-free element-

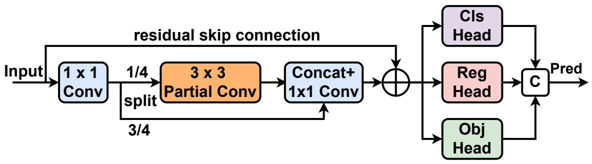

Fig. 3. Inverted Residual Decoupled Head (ISPP) with Partial Convolution (Ⓒ denotes channel concatenation).

wise maximum:

$$z_{i,c} = \max\left(z_{i,c}^{\text{sem}},\, z_{i,c}^{\text{spec}}\right), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \tag{2}$$

where taking the elementwise maximum guarantees monotonic evidence preservation ($z_{i,c} \geq z_{i,c}^{\text{sem}}$ and $z_{i,c} \geq z_{i,c}^{\text{spec}}$) at every spatial grid cell. Candidate cells with routing priority $s_i > \tau$ are clustered into bounding patches $\mathcal{P}$ and forwarded downstream.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \ldots, N_{\text{gt}}\}$. Assuming independent cell activations, the joint probability that instance j is covered by at least one selected cell is:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \tag{3}$$

The soft-coverage loss directly optimizes joint instance coverage:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log\left(p_j^{\text{cover}} + \epsilon\right), \tag{4}$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in grid cells, penalizing misses on micro targets ($a_j < 4$). The gradient with respect to cell score s_i is:

$$\frac{\partial \mathcal{L}_{\text{cover}}}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \tag{5}$$

When neighboring candidate cells have low scores ($s_k \to 0$), the product term approaches 1, generating a strong gradient to boost cell i and encouraging at least one cell to activate per instance. The overall multi-task training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Lightweight Decoupled Head and Staged Training

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an inverted residual partial-convolution (ISPP) decoupled head [15] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP @.5	Params (M)	GFLOPs	Total Recall	BPR	FPS
UAVDT Benchmark *(1280 × 1280)*						
Faster R-CNN [16]	*[TBD]*	41.5	*[TBD]*	*[TBD]*	–	*[TBD]*
RetinaNet [17]	*[TBD]*	34.0	*[TBD]*	*[TBD]*	–	*[TBD]*
ESOD [4]	0.385	35.9	**68.2**	85.17%	0.884	**117.8**
HESOD (Ours)	**0.394**	**26.0**	74.9	**88.83%**	**0.940**	106.1
Δ *vs. ESOD*	**+0.9 pp**	**-27.5%**	+9.8%	**+3.66 pp**	**+5.6 pp**	-9.9%
SeaPerson Benchmark *(2048 × 2048)*						
Faster R-CNN [16]	0.551	43.3	1546.8	67.27%	–	23.1
RetinaNet [17]	0.473	36.4	942.7	65.98%	–	28.0
ESOD [4]	0.750	35.8	**202.4**	84.42%	0.947	**85.7**
HESOD (Ours)	**0.773**	**25.9**	208.5	**87.75%**	**0.991**	80.6
Δ *vs. ESOD*	**+2.3 pp**	**-27.6%**	+3.0%	**+3.33 pp**	**+4.4 pp**	-6.0%

BPR: Bounding Patch Recall. **Bold**: best; underline: second-best (selective models).

Selector-First Staged Training. Joint training of lightweight heads perturbs shallow selector features, degrading recall. To eliminate interference, HESOD adopts a two-stage protocol: (i) *Stage 1 (Selector Learning)*: Train the Dual-Max selector with a coupled head under $\mathcal{L}_{\text{cover}}$, establishing a recall-safe priority map; and (ii) *Stage 2 (Downstream Compression)*: Freeze the trunk (backbone, dual branches, router) and fine-tune only the lightweight ISPP neck and heads. This staged decoupling preserves peak coverage (0.940 BPR on UAVDT) while slashing GFLOPs by 16.9% and parameters by 27.5%.

4. EXPERIMENTS

4.1. Experiment Settings

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280 × 1280 px, urban vehicle surveillance across 3 classes) and **SeaPerson** [3,9] (2048 × 2048 px, maritime search-and-rescue with > 85% micro targets < 32 px). Both evaluate full-resolution images without artificial cropping.

All models employ YOLOv5m in PyTorch and train for 50 epochs under selective protocols [4]. Metrics include COCO mAP@.5, Bounding Patch Recall (BPR), and scale-stratified recall across bins: *Very Tiny* ($< 16^2$), *Tiny* (16^2–32^2), and *Small* (32^2–96^2). Latency (FPS) is measured on a single NVIDIA RTX 5090 GPU at batch size 1 with FP32 precision.

4.2. Main Benchmark Comparisons

Comparison with Dense Baselines. As shown in Table 1, dense detectors incur prohibitive overhead on high-resolution grids. Compared to Faster R-CNN and RetinaNet on SeaPerson (2048 × 2048), selective detection yields massive gains of **+22.2 pp** and **+30.0 pp** mAP@.5 while slashing GFLOPs by **7.4**× and **4.5**× at **80.6** FPS, confirming the necessity of selective computation for resource-efficient high-resolution inference. The recall gap is equally stark: at a fixed matching protocol, both dense detectors miss roughly a third of all instances (67.27%/65.98% total recall vs. HESOD's **87.75%**), with the gap concentrated on Very Tiny targets (46.07%/50.05% vs. **75.71%**).

Comparison with Selective Baseline (ESOD). Against ESOD, HESOD achieves substantial instance recall recovery and parameter compression across both benchmarks (Table 1). On UAVDT (1280 × 1280), HESOD improves mAP@.5 to **0.394** (+0.9 pp) and BPR to **0.940** (+5.6 pp, rescuing +13,700 targets), while staged ISPP fine-tuning cuts parameters by **27.5%** (25.98M vs. 35.85M) at 106.1

Table 2. Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

Configuration	Detection Accuracy and Computational Efficiency						Physical-Scale Recall Breakdown				
	mAP @.5	**mAP @.5:.95**	**BPR**	**Params (M)**	**GFLOPs**	**FPS**	**Very Tiny** ($< 16^2$)	**Tiny** (16^2–32^2)	**Small** (32^2–96^2)	**Med/Lrg** ($> 96^2$)	**Total Recall**
UAVDT Benchmark (**1280 × 1280**)							(74,375)	(206,015)	(86,282)	(7,325)	(373,997)
(1) ESOD Baseline*	0.385	0.214	0.884	35.9	**68.2**	**117.8**	79.43%	84.21%	94.54%	59.78%	85.17%
(2) Semantic-only	0.384	0.217	0.906	35.9	75.0	113.8	79.13%	83.91%	94.00%	59.95%	84.82%
(3) Spectral-only	**0.396**	0.214	0.936	36.0	98.1	100.5	83.78%	87.83%	97.62%	64.68%	88.83%
(4) (3) + Channel-Pooled	<u>0.394</u>	0.209	**0.963**	35.9	99.3	97.6	**86.88%**	**92.23%**	**97.74%**	66.43%	**91.93%**
(5) Dual-Concat	0.371	0.205	0.919	35.9	83.9	107.2	79.09%	85.85%	95.85%	62.66%	86.35%
(6) Dual-Max	**0.395**	**0.218**	<u>0.940</u>	35.9	90.1	102.3	<u>85.69%</u>	<u>89.97%</u>	97.37%	<u>65.99%</u>	<u>90.36%</u>
(7) **HESOD (Ours)**	<u>0.394</u>	<u>0.215</u>	<u>0.940</u>	**26.0**	<u>74.9</u>	<u>106.1</u>	80.76%	88.98%	<u>97.11%</u>	**69.16%**	88.83%
SeaPerson Benchmark (**2048 × 2048**)							(82,417)	(174,816)	(42,987)	(155)	(300,375)
(1) ESOD Baseline*	0.750	0.320	0.947	35.8	**202.4**	**85.7**	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only	0.769	0.325	**0.991**	35.8	266.8	73.5	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only	0.770	0.327	**0.992**	35.8	267.4	71.7	75.23%	91.69%	**95.89%**	<u>86.45%</u>	87.77%
(4) (3) + Channel-Pooled	0.767	0.324	0.988	35.8	263.4	72.7	**76.13%**	91.36%	95.50%	**87.10%**	87.77%
(5) Dual-Concat	0.772	0.326	0.986	35.8	281.2	69.8	<u>76.00%</u>	91.50%	95.68%	80.00%	<u>87.84%</u>
(6) Dual-Max	**0.778**	**0.330**	**0.991**	35.8	255.1	73.8	75.65%	**92.06%**	<u>95.85%</u>	81.94%	**88.10%**
(7) **HESOD (Ours)**	<u>0.773</u>	<u>0.327</u>	**0.991**	**25.9**	<u>208.5</u>	<u>80.6</u>	75.71%	<u>91.47%</u>	95.72%	80.65%	87.75%

*Trained with standard BCE (others use $\mathcal{L}_{\mathrm{cover}}$). Parenthetical values denote ground-truth counts per bin. One-to-one matching at conf ≥ 0.001, IoU ≥ 0.5. **Bold**: best; <u>underline</u>: second-best.

FPS (+3.66 pp total recall). Recall improves across all classes: *car* ($85.6\% \rightarrow 89.0\%$), *truck* ($81.1\% \rightarrow 86.5\%$), and *bus* ($67.7\% \rightarrow 81.4\%$, **+13.7 pp**). On SeaPerson (2048 × 2048), HESOD reaches **0.773** $\mathrm{mAP}@.5$ (+2.3 pp over ESOD) and **0.991** BPR (+4.4 pp, total recall $84.42\% \rightarrow 87.75\%$), while staged ISPP compression cuts parameters by **27.6%** (25.92M vs. 35.78M) and recovers Dual-Max compute overhead down to 208.5 GFLOPs at 80.6 FPS. The Dual-Max selector reference achieves peak accuracy of **0.778** $\mathrm{mAP}@.5$ and **88.10%** recall.

4.3. Ablation Studies and Scale Analysis

Evidence Preservation Principle. As reported in Table 2, unconstrained affine concatenation (Arm 5, 0.371 mAP) degrades performance because unconstrained linear combinations dilute confident cues. In contrast, parameter-free Elementwise Max (Arm 6) enforces monotonic evidence preservation ($z_{i,c} \geq \max(z_{i,c}^{\mathrm{sem}}, z_{i,c}^{\mathrm{spec}})$), achieving peak selector accuracy of **0.395** mAP on UAVDT and **0.778** mAP on SeaPerson (0.991 BPR) while maintaining stable high-IoU localization ($\mathrm{mAP}@.5:.95$ of 0.218 and 0.330).

Efficiency Recovery via Staged Training. The inverted residual ISPP head serves as a dedicated compute compensator. On UAVDT, staged training freezes the converged selector before fine-tuning ISPP (Arm 7), preserving 0.394 $\mathrm{mAP}@.5$ and 0.940 BPR while slashing GFLOPs by 16.9% (90.1 → 74.9) and parameters by **27.5%** at 106.1 FPS. On SeaPerson, staged ISPP fine-tuning similarly cuts GFLOPs by **18.2%** (255.1 → 208.5) and parameters by **27.6%** (35.79M → 25.92M) with BPR fully preserved at 0.991, raising FPS from 73.8 to 80.6 (+9.2%) with minimal accuracy trade-off (0.778 → 0.773 mAP).

Micro-Scale Recall Recovery. In the scale breakdown of Table 2, baseline ESOD suffers severe degradation on *Very Tiny* targets ($< 16^2$ px), dropping to 79.43% on UAVDT and 74.03% on SeaPerson due to early semantic collapse. Dual-Max routing restores extreme-scale sensitivity, boosting *Very Tiny* recall by **+6.26 pp** on UAVDT and improving *Tiny* recall by **+5.28 pp** on SeaPerson (92.06% vs. 86.78%).

Channel Pooling Benefit. Comparing unpooled spectral filtering (Arm 3) with channel pooling (Arm 4) confirms that compressing channels via max/mean pooling maintains $\mathcal{O}(1)$ width complexity while improving *Very Tiny* recall by **+3.10 pp** (86.88% vs. 83.78% on UAVDT) by suppressing intra-channel noise and preventing VRAM explosion.

Error Bottleneck Diagnosis. Error analysis reveals that dual-evidence routing and coverage optimization reduce selector-dropped false negatives from 25.6% in baseline ESOD to 19.2%, shifting the remaining detection challenge from early-stage pruning to downstream localization refinement.

Latency Profiling. On an RTX 5090 GPU, stem extraction takes 1.2 ms, Dual-Evidence selection takes 1.8 ms, AdaSlicer takes 0.9 ms, and ISPP execution takes 5.5 ms, reaching 106.1 FPS on UAVDT and 80.6 FPS on SeaPerson.

Qualitative Comparison. As shown in Fig. 1, baseline ESOD misses faint micro targets due to semantic degradation, whereas HESOD's dual-evidence routing accurately localizes tiny instances without false alarms.

5. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence framework for small object detection in high-resolution vision. By coupling semantic objectness with channel-pooled spectral saliency under an evidence preservation principle and optimizing joint instance coverage via $\mathcal{L}_{\mathrm{cover}}$, HESOD prevents premature pruning on micro targets. Integrated with an inverted residual decoupled head (ISPP) fine-tuned via staged training, HESOD improves $\mathrm{mAP}@.5$ by 0.9 pp and 2.3 pp (boosting total recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, with $> 27\%$ fewer parameters and negligible FLOPs overhead, establishing an effective accuracy-efficiency co-design suitable for resource-constrained deployment. Future work will extend dual-evidence selective routing to multimodal aerial perception platforms.

6. REFERENCES

[1] Pengfei Zhu, Longyin Wen, Dawei Du, Xiao Bian, Heng Fan, Qinghua Hu, and Haibin Ling, "Detection and tracking meets drones: A benchmark," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7388–7404, 2021.

[2] Dawei Du, Yuankai Qi, Hongyang Yu, Yifan Yang, Kaiwen Duan, Guorong Li, Weigang Zhang, Qingming Huang, and Ling Shao, "The unmanned aerial vehicle benchmark: Object detection and tracking," in *European Conference on Computer Vision*, 2018, pp. 370–386.

[3] Xue-Yang Zhang, Guanying Xu, Jin-Yu Fan, and Haibin Ling, "Scale match for tiny person detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2020, pp. 1256–1265.

[4] Kai Liu, Zhunga Fu, Shan Jin, Zhipeng Chen, Feng Zhou, Rong Jiang, Yonghong Chen, and Jinwei Ye, "ESOD: Efficient small object detection on high-resolution images," *IEEE Transactions on Image Processing*, vol. 34, pp. 183–195, 2025.

[5] Mahyar Najibi, Bharat Singh, and Larry S. Davis, "AutoFocus: Efficient multi-scale inference," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 9745–9755.

[6] Chenhongyi Yang, Zehao Huang, and Naiyan Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16668–16677.

[7] Bowei Du, Yanan Huang, Jiahui Chen, and Dong Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13435–13444.

[8] Hao Sun, Rui Wang, Yufeng Li, Chenhongyi Yang, Zehao Huang, Naiyan Wang, and Lin Zhou, "SET: Spectral enhancement for tiny object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 1–10.

[9] Leon A. Varga, Benjamin Kiefer, Martin Messmer, and Andreas Zell, "SeaDronesSee: A maritime benchmark for detecting humans in open water," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 2260–2270.

[10] Fanyi Yang, Heng Fan, Peng Chu, Erik Blasch, and Haibin Ling, "Clustered object detection in aerial images," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 8311–8320.

[11] Changlin Li, Taojiannan Tao, Chen Gong, Xiaojun Chang, Liang-Chieh Chen, and Zizhang Wu, "Density map guided object detection in aerial images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 190–199.

[12] Jinwang Wang, Chang Xu, Wen Yang, and Lei Yu, "A normalized gaussian Wasserstein distance for tiny object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2021.

[13] Chang Xu, Jinwang Wang, Wen Yang, Huai Yu, Lei Yu, and Gui-Song Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *European Conference on Computer Vision*, 2022, pp. 526–543.

[14] Chen Guo, Yang Li, Jiaman Ma, Zhenyu Fang, Zuoang Niu, and Hongbo Xu, "RESBIDET: Efficient dual-branch small object detection for UAVs under resource-constrained conditions," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 3156–3160.

[15] Xiaocong Xu, Qi Li, Jeng-Shyang Pan, Linfeng Wei, and Xiao-Zhi Gao, "ESOD-YOLO: An enhanced efficient small object detection framework for aerial images," *Computing*, vol. 107, pp. 54, 2025.

[16] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.

[17] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2017, pp. 2980–2988.